

Austin Animal Shelter Data Analysis and Classification

Overview

This project analyzes outcomes of animals at the Austin Animal Shelter. Using a dataset of over 131,000, classification models were applied to predict whether an animal was adopted or transferred. Data cleaning, feature engineering, and exploratory analysis were also conducted before using the classification models.

Data Preparation

The raw dataset contained 131,165 rows and 12 columns. Initial checks showed that all variables were stored as object types, including dates and numerical values such as “Age upon Outcome.” Because of this, conversions were necessary for the following:

- **“DateTime”** and **“Date of Birth”** → converted to datetime objects.
- **“Age upon Outcome”** → converted into a continuous variable in days for consistency.

17 duplicate rows were found, which were removed. Missing values were handled as follows:

- **“Name”**: replaced with "No Name" for missing name value, since that is a common occurrence
- **“Outcome Type”**: only 40 missing values, dropped to avoid ambiguity.
- **“Outcome Subtype”**: had over 65,000 missing entries, was dropped as irrelevant for prediction.

After cleaning, non-predictive columns (Animal ID, Date of Birth, Name, DateTime, Outcome Subtype) were also dropped.

To handle “Breed” and “Color”, as they are categorical variables with very high cardinality, they were reduced to the top 20 breeds and top 15 colors. The rest were grouped as “Other”. Finally, one-hot encoding expanded the feature set to 183 variables.

Note: Part 2 states to drop all “Breed” columns, so the dataset used for the predictive models has 162 features.

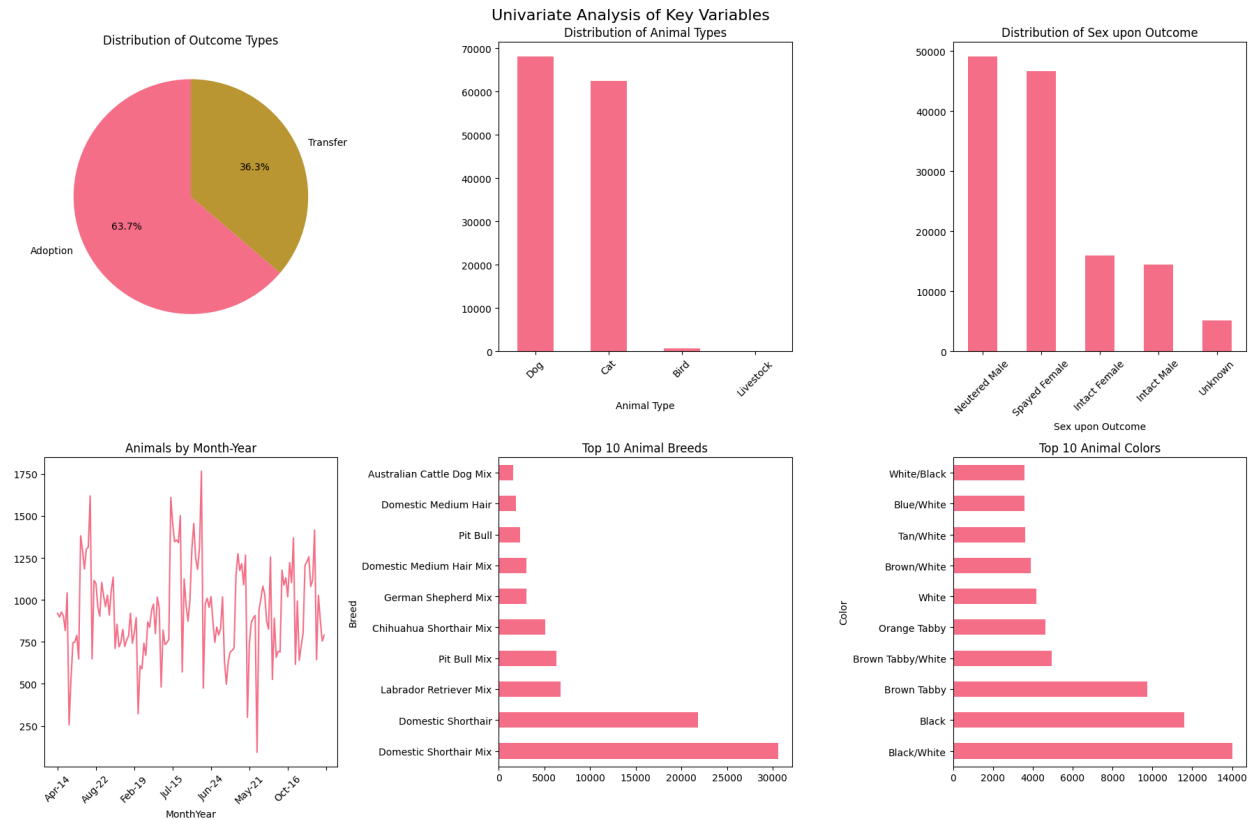


Figure 1: Univariate Analysis of Key Features

Insights from Data Preparation

Exploratory analysis revealed:

- **Adoption** dominated outcomes (64%), with Transfers at 36%, creating a class imbalance.
- **Dogs** were the most common species, and **Neutered Males** the most common sex category.
- The median animal age was **~244 days**, but the distribution was skewed by much older outliers.
- Seasonal patterns were visible in the **MonthYear** variable, with intake surging in certain months of the year.

These insights guided feature engineering and emphasized the need for careful metric selection in modeling, since imbalance meant that using accuracy alone could be misleading.

Modeling Procedure

The data was split into **80% training** and **20% testing**, using **stratified sampling** to preserve outcome class ratios (Adoption and Transfers). Features were standardized for models sensitive to scaling, K-Nearest Neighbor, and Logistic Regression.

Three classifiers were used:

1. **Basic KNN (k=5)**: a straightforward baseline.
2. **KNN with Grid Search 5-fold CV**: hyperparameter tuning ($k \in \{3, 5, 7, 9, 11\}$)
3. **Perceptron (linear model)**: reliable, easy to interpret, and efficient with large, sparse datasets.

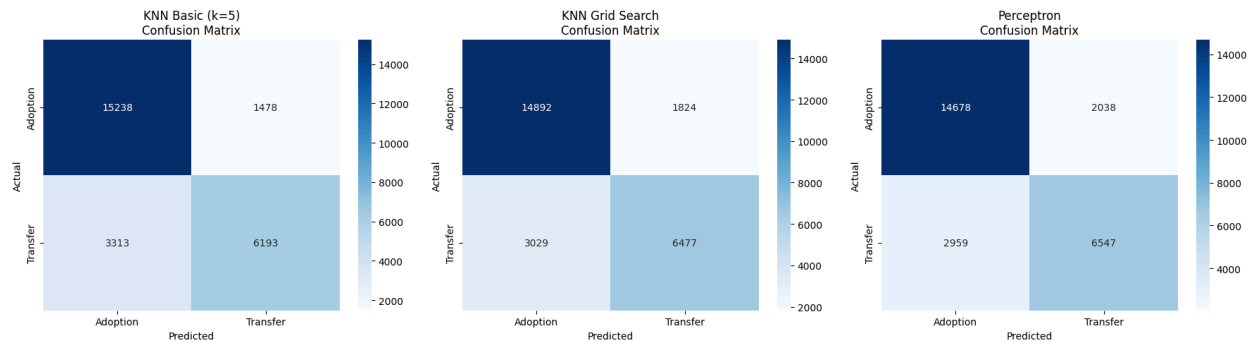


Figure 2: Confusion Matrices for KNN ($k=5$), KNN with Grid Search ($k=3$), and Perceptron

Model Performance

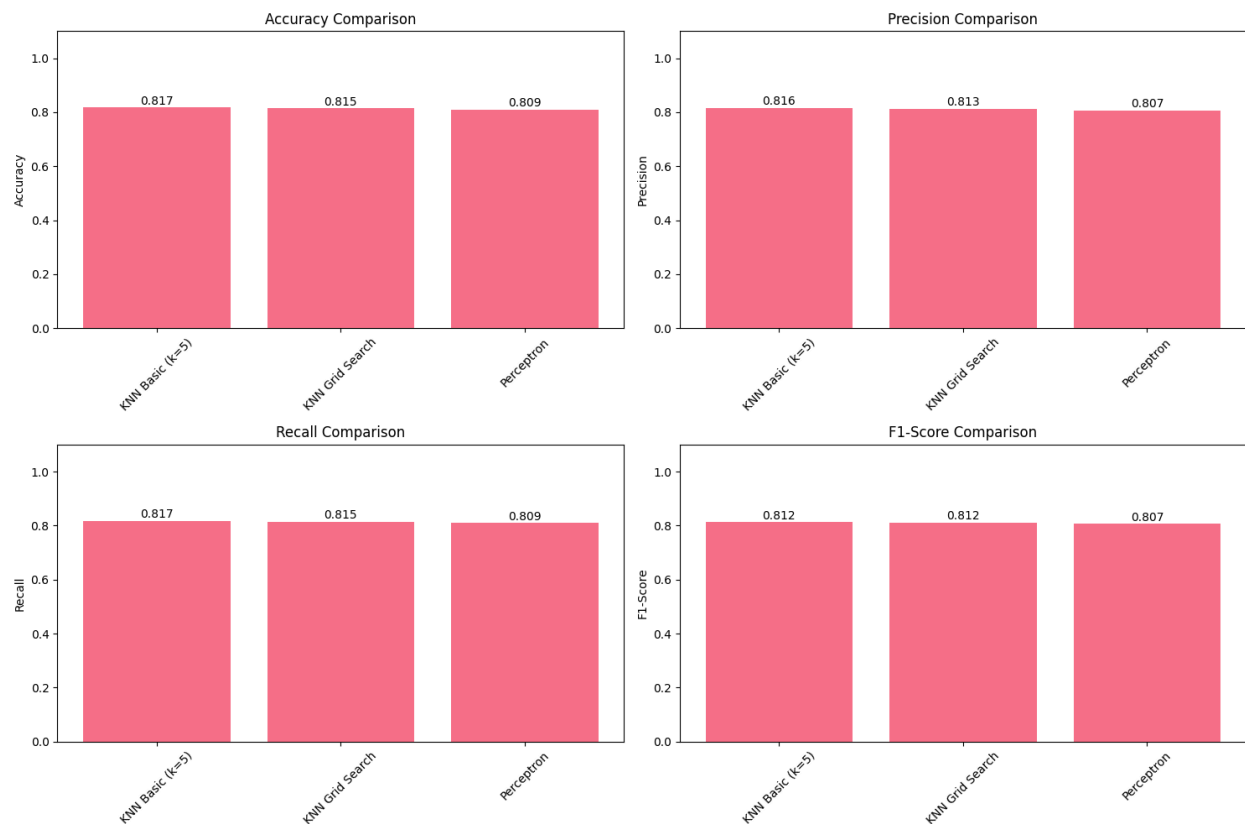


Figure 3: Model Performance Comparison—Accuracy, Precision, Recall, F1-Score for all three models

The results showed clear differences:

- **KN- Fixed (k=5):** Accuracy 81.7%, strong recall for adoptions but weaker on transfers.
- **KNN (Grid Search, best k=3):** Accuracy 81.5%, slightly improved f1 for Transfers, but no overall gain.
 - Note: Accuracy was chosen as the primary evaluation metric due to the relatively balanced dataset.
- **Perceptron:** Accuracy 80.94%, with the overall lowest performance.

Overall, the KNN Basic classifier performed slightly better than the KNN-Grid Search and Perceptron in all major metrics.

Most Important Metric

For this problem, **accuracy** is the most important evaluation metric. The dataset is only moderately imbalanced, so using the F1-score to correct for imbalance is less critical. Accuracy provides a clear measure of overall predictive performance, reflecting how well the model classifies both adoption and transfer outcomes in aggregate. Accuracy was chosen over precision and recall because neither outcome class is rare enough to demand prioritizing one type of error. Precision would emphasize avoiding false adoptions, and recall would emphasize catching every transfer, but prioritizing only one could bias the model away from balanced overall performance. Since both outcomes are meaningful and occur at substantial rates, accuracy gives the fairest assessment of how well the classifier works across the entire dataset.

Confidence in the Model

I am moderately confident in the KNN-Fixed model. With over 130k rows and balanced cross-validation, the results are statistically reliable. However, some caution is necessary as:

- The dataset only contains outcomes from one shelter (Austin), which may limit generalization to other contexts.
- Feature reduction (ex, removing “Breeds” and collapsing “colors”), though simplified modeling, likely reduced the granularity of the dataset, which directly affects the predictive model.

Final Recommendation: The KNN-Fixed, $K = 5$ should be used for predicting adoption versus transfer outcomes, with the understanding that further refinement could yield even stronger results.

Conclusion

Through systematic data preparation, treatment of missing values, and encoding of categorical variables, we created a dataset suitable for classification tasks. Univariate analysis provided key context, showing that adoptions were the majority outcome, dogs were the most common species, and neutered males were the most common sex category. These patterns revealed the class imbalance in the data and informed our choice of evaluation metrics. Comparing multiple models, the K-Nearest-Neighbor-Fixed approach achieved the best balance of accuracy, precision, recall, and f1-score.